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1  PROGRAM Controller_ST
2  VAR
3      Active_Recipe : INT ;
4      Current_Bin   : INT ;
5      Target_Accum  : REAL ;
6      Recipe_Total  : REAL ;
7      Complete_Hold : BOOL ;
8      Start_Edge    : R_TRIG ;
9      Stop_Edge     : R_TRIG ;
10     Reset_Edge    : R_TRIG ;
11     Complete_Timer : TON ;
12     Stop_Requested : BOOL ;
13 END_VAR
14
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```
1  Start_Edge ( CLK := Start ) ;
2  Stop_Edge ( CLK := Stop ) ;
3  Reset_Edge ( CLK := Alarm_Reset ) ;
4  Manual_Discharge_Edge ( CLK := Manual_Discharge ) ;
5
6  Alarm_Active := Alarm_Underweight OR Alarm_EmptyBin ;
7
8  IF Stop_Edge.Q AND Batch_State >= 1 AND Batch_State <= 4 THEN
9      Stop_Requested := TRUE ;
10 END_IF
11
12 // E-Stop → FAULT (overrides everything)
13 IF E_Stop AND Batch_State <> 5 THEN
14     Batch_State := 5 ;
15 END_IF
16
17
18 CASE Batch_State OF
19     0 : // IDLE
20         IF Start_Edge.Q AND Weight_LoadCell < 5.0 AND NOT Alarm_Active
21             THEN
22                 Active_Recipe := Recipe_Select ;
23                 Recipe_Total := Recipe_Table [ Active_Recipe , 1 ] + Recipe_Table [
24                     Active_Recipe , 2 ]
25                     + Recipe_Table [ Active_Recipe , 3 ] + Recipe_Table [
26                     Active_Recipe , 4 ] ;
27                 Target_Accum := 0.0 ;
28                 Current_Bin := 1 ;
29                 Batch_State := 1 ;
30             END_IF
31         1 : // DOSING
32             Current_Target := Target_Accum + Recipe_Table [ Active_Recipe ,
33                 Current_Bin ] ;
34             CASE Current_Bin OF
35                 1 :
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33      IF BinLevel_1 < EMPTY_THRESHOLD THEN
34          Alarm_EmptyBin := TRUE ;
35          Batch_State := 5 ;
36      ELSIF Weight_LoadCell >= Current_Target THEN
37          Gate1_Activate := FALSE ;
38          IF Gate1_Closed THEN
39              Target_Accum := Current_Target ;
40              Current_Bin := 2 ;
41          END_IF
42      ELSE
43          Gate1_Activate := TRUE ;
44      END_IF
45  2 :
46      IF BinLevel_2 < EMPTY_THRESHOLD THEN
47          Alarm_EmptyBin := TRUE ;
48          Batch_State := 5 ;
49      ELSIF Weight_LoadCell >= Current_Target THEN
50          Gate2_Activate := FALSE ;
51          IF Gate2_Closed THEN
52              Target_Accum := Current_Target ;
53              Current_Bin := 3 ;
54          END_IF
55      ELSE
56          Gate2_Activate := TRUE ;
57      END_IF
58  3 :
59      IF BinLevel_3 < EMPTY_THRESHOLD THEN
60          Alarm_EmptyBin := TRUE ;
61          Batch_State := 5 ;
62      ELSIF Weight_LoadCell >= Current_Target THEN
63          Gate3_Activate := FALSE ;
64          IF Gate3_Closed THEN
65              Target_Accum := Current_Target ;
66              Current_Bin := 4 ;
67          END_IF
68      ELSE
69          Gate3_Activate := TRUE ;
70      END_IF
71  4 :
72      IF BinLevel_4 < EMPTY_THRESHOLD THEN
73          Alarm_EmptyBin := TRUE ;
74          Batch_State := 5 ;
75      ELSIF Weight_LoadCell >= Current_Target THEN
76          Gate4_Activate := FALSE ;
77          IF Gate4_Closed THEN
78              IF ABS ( Weight_LoadCell - Recipe_Total ) >
Recipe_Total * TOLERANCE THEN
79                  Alarm_Underweight := TRUE ;
80                  Batch_State := 5 ;
81              ELSE
82                  Batch_State := 2 ;
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83             END_IF
84         END_IF
85     ELSE
86         Gate4_Activate := TRUE ;
87     END_IF
88 END_CASE
89
90 2 : // MIXING
91     Mixer_Activate := TRUE ;
92     IF Mixer_Done THEN
93         Batch_State := 3 ;
94     END_IF
95
96 3 : // DISCHARGING
97     Mixer_Activate := TRUE ;
98     Discharge_Activate := TRUE ;
99     IF Weight_LoadCell < 5.0 THEN
100         Discharge_Activate := FALSE ;
101         Batch_State := 4 ;
102     END_IF
103
104 4 : // COMPLETE
105     Complete_Timer ( IN := Complete_Hold , PT := T#2S ) ;
106     IF NOT Complete_Hold THEN
107         Batch_Counter := Batch_Counter + 1 ;
108         Total_Weight := Total_Weight + Recipe_Total ;
109         Complete_Hold := TRUE ;
110     END_IF
111     IF Complete_Timer . Q THEN
112         Complete_Hold := FALSE ;
113         IF Stop_Requested OR ( Target_Batches > 0 AND Batch_Counter >=
Target_Batches ) THEN
114             Stop_Requested := FALSE ;
115             Batch_State := 0 ;
116             ELSIF Auto_Restart THEN
117                 Target_Accum := 0.0 ;
118                 Current_Bin := 1 ;
119                 Batch_State := 1 ;
120             ELSE
121                 Batch_State := 0 ;
122             END_IF
123         END_IF
124
125 5 : // FAULT
126     Gate1_Activate := FALSE ;
127     Gate2_Activate := FALSE ;
128     Gate3_Activate := FALSE ;
129     Gate4_Activate := FALSE ;
130     Mixer_Activate := FALSE ;
131     Discharge_Activate := FALSE ;
132     IF Reset_Edge . Q THEN

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133         Alarm_Underweight := FALSE ;
134         Alarm_EmptyBin := FALSE ;
135         Batch_State := 0 ;
136     END_IF
137 END_CASE
138
139 // After CASE - manual discharge logic:
140 IF Manual_Discharge_Edge .Q AND ( Batch_State = 0 OR Batch_State = 5 ) THEN
141     Manual_Discharge_Latch := NOT Manual_Discharge_Latch ;    // toggle: press =
142     ON, press again = OFF
143 END_IF
144 IF Manual_Discharge_Latch AND Weight_LoadCell <= 5.0 THEN
145     Manual_Discharge_Latch := FALSE ;    // auto-stop when empty
146 END_IF
```